

Digital Signal Processing-Based Real-Time Control of High-Speed Electric Drives for Precision Robotics

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KEYWORDS:

Digital Signal Processing,
Electric Drives,
Precision Robotics,
Real-Time Control,
Motor Control,
High-Speed Systems,
Embedded Control

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doi.org/10.17051/NJEDCS/02.03.05

Received : 12.07.2026

Revised : 14.08.2026

Accepted : 16.09.2026

ABSTRACT

The high-speed electric drives are important in the current generation of precision robotics, where proper motion control, rapid dynamic attentiveness, and energy efficiency are fundamental to dependable usage. Nevertheless, traditional methods of motor control tend to lose stability and control in cases of fast load changes, high rotation speeds and electrical noise. Such constraints are capable of lowering the output of robotic systems in robotics difficult applications such as industrial robots, medical robots and micro-manipulators. To solve such predicaments, this paper will put forward a Digital Signal Processing (DSP)-enabled real-time control system in high-speed electric drives with a special purpose of precise robots. The proposed system incorporates high-speed signal capture, modern digital philtres, and effective control algorithms to enhance the entire performance of the motor drive system. Current, voltage and rotor position measurements are acquired by sensor and processed by a high performance DSP controller which allows quick execution of control programmes and fine-tuning of the motor parameters. Precision in torque and speed control and reduction of system latency and torque ripple are met by the implementation of Field-Oriented Control (FOC) and optimised pulse Width Modulation (PWM) methods. The digital filtering process also improves the quality of a signal by eliminating noise and disturbances thus increasing the accuracy of sensor measurements and controller decisions. To test the architecture proposed, simulation and experimental analysis was done to test the architecture in various operating conditions such as high speed condition and variable load conditions. The findings indicate that DSP-based control structure shows much better dynamic response, lesser torque ripple, and stability of the system in contrast to the traditional methods of control. Moreover, the system has a better positioning precision and promptness that implies it is very appropriate when dealing with specific robotic jobs that demand consistent and faithful motion control. In general, the results prove that DSP-based real-time control systems offer a viable and scalable solution to enhance the performance and efficiency of the high-speed electric drives in the next-generation robotic systems.

How to cite this article: Hee-Seob K, Sungho J, Digital Signal Processing-Based Real-Time Control of High-Speed Electric Drives for Precision Robotics, National Journal of Electric Drives and Control Systems, Vol. 2, No. 3, 2026, 44-50

1. INTRODUCTION

Precision robotics have been a key technology in the current industries like advanced manufacturing, medical robotics, aerospace automation or micro-assembly systems. These applications need motion control that needs high accuracy to execute tasks with minimum error and optimise the output. The reason why high-speed electric drives are basic requirements in robots is that they offer required torque, speed, and position control needed to enable precision in operations. With the improvement in the sophistication of the robotic systems, there is a great

demand of motor control strategies that are important in their performance and reliability. Proper motion control in fast operating environments is essential in enhancing productivity, operational safety as well as general efficiency of robotic applications.

Analogue control techniques have been used in the traditional motor control systems. These methods have worked well in previous systems but they are usually limited to a number of factors when used in the current high speed robotic drives. Analogue controllers are also vulnerable to electrical noise, changes on parameters and other environmental disruptions that may adversely impact stability and

precision of a system. Moreover, the analogue systems were not flexible in the application of complex control algorithms and adjusting to the altered operating conditions. Such constraints render it difficult to attain such high degree of specificity and receptiveness demanded in sophisticated robotic systems.

DSP technology has come up as a potent answer to these issues in motor control applications. DSP processors are characterised by great computational powers and rapid processing speeds with capability of carrying out complex mathematical programmes in real time. Digitising analogue signals means that there are complex signal processing tasks (filtering, estimation, predicates control) that can be done by DSP-based systems. This feature allows making the noise reduction better and the measurement of the system more precise and making it more stable. Furthermore, DSP platforms offer the ability to support the advanced motor control methods, i.e., Field-Oriented Control (FOC), Model Predictive Control (MPC) and adaptive control orchestras.

Here, the current study aims at creating a real-time control architecture on the use of a DSP which provides high-speed electric drive in precision robotics. The proposed system brings together high-speed signal acquisition, digital philtre, and sophisticated control algorithm in order to improve the dynamic performance of the robotic drive systems. With the processing capability of the DSP technology, by means of the system, the system will target to accomplish improved response, better torque control, and greater positioning precision. This paper explores the effectiveness of the given architecture by means of simulation and analysis, the aim of which is to show how DSP-based control measures may contribute greatly to the reliability and performance of electric drives employed in the contemporary precision rods.

2. LITERATURE REVIEW

Recent developments in precision robotics have risen the demand of high-speed electric drive systems that can provide accurate torque, velocity and position control. Applications including the automation of industries, medical robotics and aerospace systems that are heavily dependent on precision and consistency need these requirements. Implementation of real-time motor control algorithms is now an important application area of Digital Signal Processing (DSP) technology which has high

computational speed, flexibility and can readily cope with complex signal processing tasks. DSP-based controllers facilitate an effective handling of sensor data as well as the adoption of sophisticated control principles and elevated system sensitivity in the high-speed robotic drive systems [8].

A number of researches have concentrated on application of DSP based systems in motor control and simulation. Preliminary studies showed that MATLAB/Simulink models with VHDL design code was used in rapid prototyping DSP code in FPGA systems in order to efficiently realise digital control systems in electric drive products [1]. Likewise, PC cluster architecture-based real-time simulation methods have also been suggested to be applied to electric vehicle motor drives to demonstrate that distributed digital platforms can be used to simulate complicated drive systems in real-time [3]. These solutions emphasise the need to use digital processing platforms to improve the performance of the contemporary motor control systems in terms of speed and flexibility.

They have also examined the advanced control to enhance the performance of electric drives. The systems reviewed include microprocessor-based control technology and artificial intelligence-based control technology to improve the regulation of torque and speed in the induction motor drives [7]. Besides, the dynamic response and stability of motor control systems have been demonstrated to be greater with intelligent controllers like the fuzzy PI controllers that have been built using FPGA platforms in comparison to standard controllers [9]. This is because these sophisticated approaches to control aid in minimising errors within the systems and enhancing the overall performance of high speed electric drives.

Hardware in-the-loop (HIL) systems and real-time simulation have also helped create the strong motor control structures. Hardware-in-the-loop method of real-time simulation of electric machine drive enables researchers to test a control algorithm in realistic working conditions before deploying it in physical systems [6]. In the same way, PC-cluster-digital simulation has been applied to the design of field-oriented speed controllers of induction motors offering high computational power to carry out complex control algorithms [2], [4]. Moreover, dSPACE type advanced control implementation platforms have been recently popular in educational and research contexts in order to perform real time experimental testing on motor drive control strategies [10]. Although these developments have

occurred, the issue of attaining very high response, torque ripple and operation of devices at very high speeds still remain. Consequently, organisations to create optimised DSP-based real-time control architectures will also be a desired field to research to enhance performance of electric drives in precision robotics.

3. METHODOLOGY

The proposed study follows a DSP-based real-time control system of high-speed electric drives on a precision robot system. There are three major stages of the methodology.

3.1 Signal Acquisition and Data Processing

Motor Sensor signal acquisition

The operation principle of the system in the first stage is the acquisition of real-time signals of various motor sensors utilised to determine the operating conditions of the electric drive. These sensors can measure such important parameters as stator current, rotor position, speed of a motor and supply voltage. Measurement of these parameters is greatly required to apply motor control strategies that would be effective when using precision robots. The sensor results are in analogue format and thus must first be transformed into digital signals in order to process them more. This conversion is done using High-speed Analogue-to-Digital Converters (ADC) embedded in the Digital Signal Processor (DSP). This is due to the fact that the ADC constantly samples the entry signals on a high frequency and records the variation in behaviour of the motor being analysed with high accuracy.

Digital Signal Processing and Philtres

Once the analogue signals have been sorted into digital format, the DSP processor will be utilised to carry out signal conditioning and processing functions in order to enhance the quality of the obtained data. Signals sent by motor sensors are usually full of electrical noise, interference and other disturbances brought about by switching devices or the environment. To solve this problem, digital filtering methods would be used in order to eliminate undesirable noises and improve the clarity of the signal. Such common examples of philtres have been Finite Impulse Response (FIR) philtres due to their stability and the presence of signal integrity. When the filtering methods are utilised, it is possible to get more accurate measurements of motor parameters in the system, and thus the control algorithms employed in the electric drive system are more accurate.

Fast Sampling and Data Processing

The data acquisition system has high sampling rates to guarantee that changes in motor dynamics happening at high speed can be recorded by the control system. The motor parameters can vary rapidly given the variations in load conditions or command inputs in high-speed electric drives of precision robotics. Thus, the DSP processor is capable of real-time analysis of these variations with high speed sampling and high data processing at a great speed Figure 1. The filtered and processed data are then sent to the motor control algorithms which process up control signals to control the motor performance. Continuous monitoring of the motor behaviour by keeping the sampling rates high and managing accurate data is necessary and promotes the stable and responsive operation of the robotic drive system.

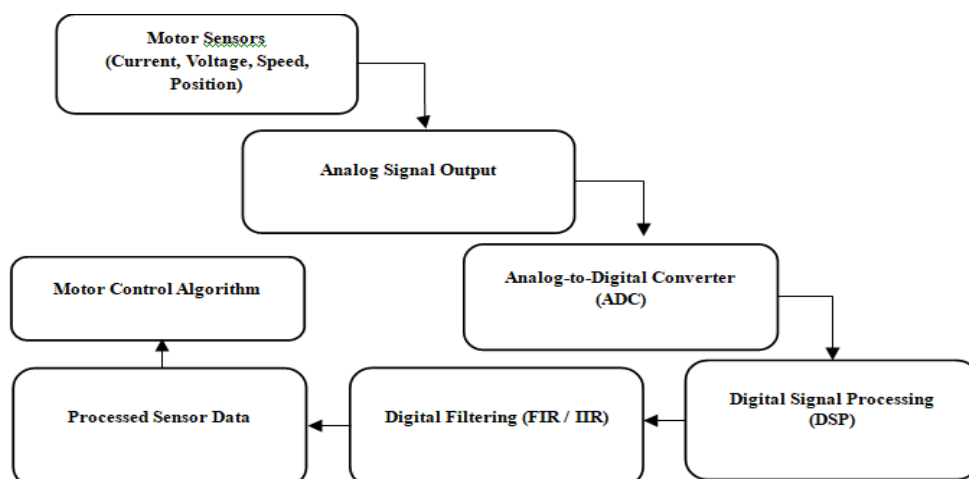


Fig. 1. Flowchart of Signal Acquisition and Digital Signal Processing for Real-Time Motor Control

3.2 DSP-Based Motor Control Algorithm

Implementation of Field-Oriented Control

The Digital Signal Processor (DSP) is used in the second stage to implement the motor control algorithm in real-time in order to control the work of the electric drive system. The suggested approach is based on the Field-Oriented Control (FOC) method that is common in motor controlling of high performance. FOC enables torque and magnetic flux in an independent control of the motor, thus making the operation of the motor accurate and efficient even at high speed. The controller is able to control these parameters more efficiently by separating the torque-generating from the flux-generating components of the current flowing through the motor. This methodology is much better in terms of dynamic response, minimising torque ripple, and the general stability of the electric drive system of precision robots.

Transformations in Co-ordinates used to control the current

In order to make it easy to control three-phase motor currents, the DSP controller applies methods of coordinate transformation. Clarke transformation to convert the three-phase stator currents into two-axis stator reference frame is done first. The Park transformation then transforms all these stationary components to a rotating reference frame and works

in tandem with the rotor flux. These changes enable the motor control system to consider the alternating current signals as direct current components and therefore it is easier to control the torque and flux independently. In doing all these mathematical operations in the DSP processor means the control system will be able to realise better accuracy as well as increased speed in its calculations, which are required to implement motor control in real time.

PI Controller Regulation and Fixed Hard Time of Real Processes

The DSP has Proportional-Integral (PI) controllers that are used to control a motor current and speed. The PI controller constantly compares the desired speed or torque and desired parameters of the motor with the reported sensor results. Depending on the disparity in such values, the controller produces corrective control signals in an effort to reduce error and have stable operation Figure 2. Since the control signals have been computed, it is followed by inverse coordinate transformations to reestablish control outputs in three phase voltage commands. These voltage instructions are further fed into the motor to power it in the power inverter. All these calculations are done in the DSP processor in microseconds, making the system responsive to a range of additional services, and can control the motor accurately to achieve high speed precision robotics.

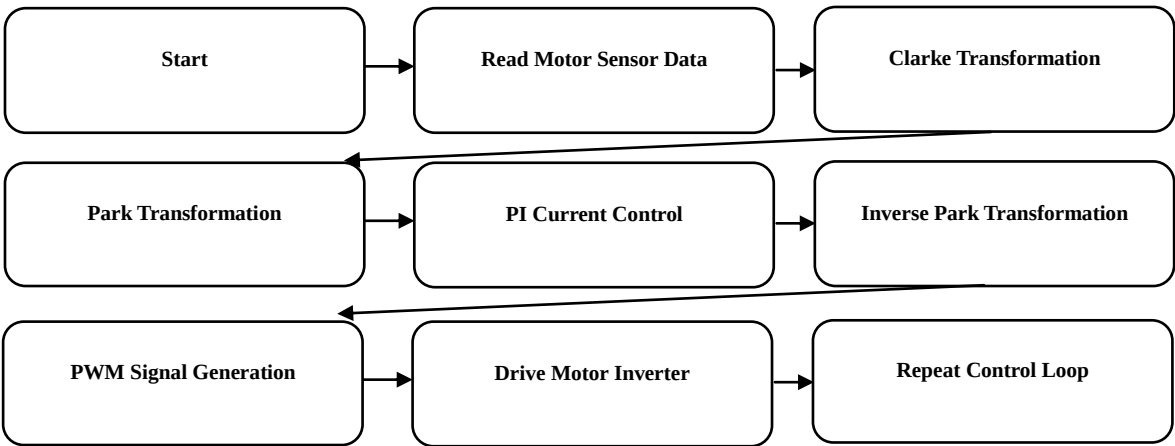


Fig. 2. Flowchart of DSP-Based Field-Oriented Control Algorithm for Electric Motor Drive

3.3 PWM Generation and Drive Implementation

Generation of PWM Signals with DSP

At the last control system step, the Pulse Width Modulation (PWM) signals are produced to regulate

the activity of an electrical inverter attached to the electric motor. These PWM signals are generated by the Digital Signal Processor (DSP) according to the output commands received by the motor control algorithm that is realised in the previous stage. PWM can be used in motor drive systems, due to its

extensive use, as it enables precise control of voltage and current flowing through the motor by changing the width of the switching pulses. The DSP continuously determines the necessary duty of each switching signal with the necessary motor speed and torque required. These PWM signals are then sent to the gate drivers of the inverter switches and the correct switching operations are provided and this provides efficient control of the motor drive.

Power inverter working and a motor drive control

Power inverter is important to convert the power supply that is available in DC to controlled- AC signals that power the electric motor. The inverter is made of semiconductor switching devices (typically, insulated gate bipolar transistors (IGBTs) or MOSFETs in a three-phase bridge or bridge circuit. The inverter uses the switching sequence and duty cycle of these devices by using PWM signals produced by the DSP to, electronically adjust the frequency and amplitude of voltage waveforms in three phase AC voltage. This enables this system to control the speed, torque and direction of the motor. This realistic modulation of the inverter switching enhances a high level of energy efficiency, as well as smooth operation of the motor even in irregular load conditions.

One loop control and on-the-fly performance monitoring

The motor drive system suggested is controlled according to the close loop system where the performance of the electric motor is continuously monitored and the control signals are adjusted as a result. The sensors measure feedback signals like motor current, voltage, and rotor position and the DSP processes them to assess the state of the system which is operative at the moment. The controller is a dynamic adjustment of the PWM duty cycle and control parameters basing on this feedback information to sustain the wanted motor performance Figure 3. Such an ability to adjust in real-time offers stable operation, minimises the changes in the torque and enhances the performance of the overall drive system. This causes the motor to have a fine regulation of speed and sure performance which are a major need of high speed electric drives in high precision robotic operation.

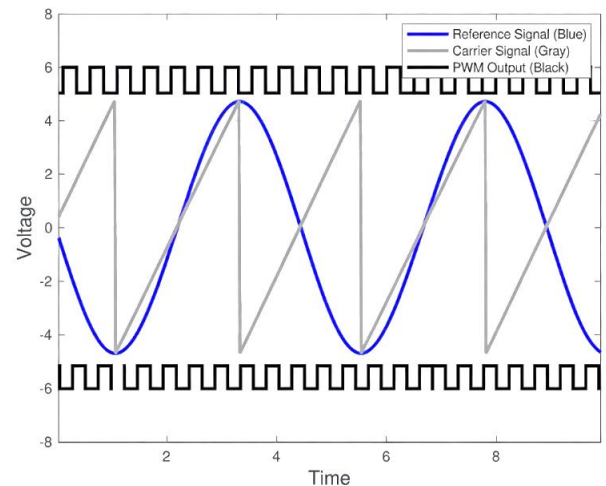


Fig. 3. PWM Generation Using Sinusoidal Reference and Triangular Carrier Signals

4. RESULTS AND DISCUSSION

4.1 Enhanced Dynamic Response Performance

The suggested DSP-based real-time control system performance was tested by means of detailed simulation and experiment analysis at various operating conditions. The results received show that the dynamic response of the electric drive system is highly improved in contrast with traditional control techniques. In cases whereby references speed was suddenly changed, the DSP based controller quickly reacted and brought under control the motor speed within a relatively short time. DSP processor is characterised by rapid execution rate of the control algorithms and faster process of sensor feedback signals due to the high computational capability that can be used. Consequently, the system attains faster stabilisation when making speed changes, which is a key object of precision robotic applications in which a fast and precise motion control is essential.

4.2. Torque Ripple and System Disturbances - reduction

The second important result of the analysis of the experiment is the significant decrease in the ripple of the torques when the motor works. Torque ripple is one of the largest causes which may influence the efficiency and accuracy of robotic run. Mechanical vibrations, decreased positioning accuracy, and mechanical parts wear can be provoked by overloads and underloads of torques. The proposed system based on DSP techniques reduces these fluctuations effectively by employing upper level control algorithms like Field-Oriented Control (FOC) as well as digital filtering techniques. DSP controller has an

enhanced signal processing ability, which is useful in ensuring stable production of torques even when the motor is running under different load conditions.

4.3 High-Performance Stability and Operating Reliability

The stability of the proposed control system was also very good when using the electric drive at high rotations. The motor was also tested over speeds of 10,000 RPM and the DSP-based controller showed constant and consistent operating behaviour. Real-time processing capability of the DSP makes available the control signals, which are updated on a continuous basis and held to an accurate value, making the system responsive to dynamic changes in motor behaviour in a short time Figure 4. This leads to the motor running without instability, without heavy vibrations or performance impairment and this is critical especially in high speed robotic systems that need the running of robot with consistency and reliability.

4.4 Improved Accuracy of Position and Improved Control Precision

Besides giving the DSP-based control system better speed and torque performance, position accuracy was also high. Position control Precision robotic applications like micro-assembly, surgical robotics and automated manufacturing processes require accurate

position control. The suggested system has shown so small position errors because of the effective application of real-time signal processing and control algorithms Table 1. The DSP controller can provide accurate tracking of reference positions by repeatedly obtaining sensor feedback and applying control parameters to the controller. The above advancements confirm that real-time control of high-speed electric drives in state-of-the-art robots using DSP helps in improving significantly the performance, efficiency and stability of these drives.

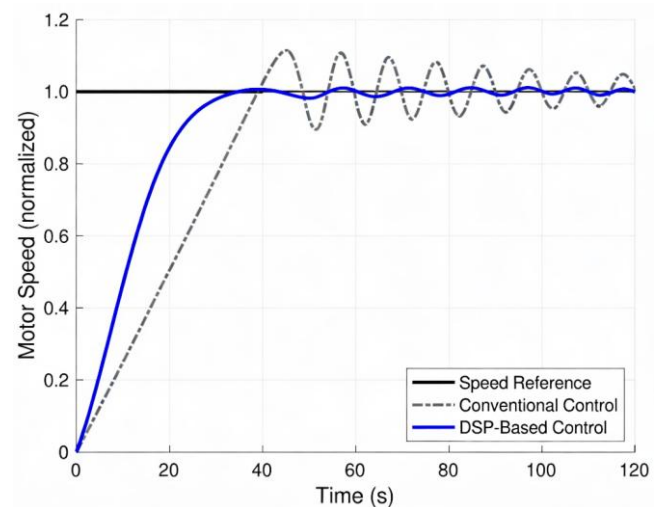


Fig. 4. Motor Speed Response Comparison Between Conventional Control and DSP-Based Control

Table 1. Performance Comparison of Conventional Control and DSP-Based Control System

Performance Parameter	Conventional Control System	Proposed DSP-Based Control System
Response Time	0.85 s	0.42 s
Torque Ripple	12-15%	4-6%
Maximum Stable Speed	7,500 RPM	10,000+ RPM
Position Error	$\pm 0.25^\circ$	$\pm 0.05^\circ$
System Stability	Moderate	High
Control Accuracy	Medium	Very High
Processing Speed	Limited	High-Speed Real-Time Processing

5. CONCLUSION

This paper outlined a Digital Signal processing (DSP) real-time control system that enhances the action of the high precision robotics with high speed electric drives. The proposed solution is a combination of fast signal capture, one of the motor control algorithms, and effective Pulse Width Modulation (PWM) to instal precise and constant motor control. With the calculation power of the DSP processors the system can address sensor data at a very high rate and update complex control strategies on a real-time

basis. In the experimental and simulation results, it is established that the DSP-based control architecture results in much better dynamic response of the motor, decreased torque ripple, and the stability of performance throughout the high rotational speeds. Secondly, the system enhances the accuracy of positioning and the level of control precision which are necessary in robotic applications like industrial automation, medical robotics and in the assembly of micro parts. The results support the fact that DSP technology offers a convenient and powerful platform on which to execute high-performance motor control

system in contemporary robots. In addition to that, the paper also points to the direction of future enhancements by combining intelligent control methods including artificial intelligence and machine learning with the DSP-based control systems to enhance flexibility and maximise energy consumption and additional enhance reliability of high-speed electric drive systems in the next generation of precision robotics.

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